Chao Ding

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RESEARCH INTERESTS

•Medical AI, Machine Learning, Artificial Intelligence, LLM, Agent

Publications and Conference Presentations

** Equal contribution / co-first authorship.*

1. **Ding, C.*,** et al.
   Conversational Fluency Does Not Ensure Clinical Safety in Patient-Facing Medical AI. Manuscript under review at *NEJM AI.*
2. **Ding, C.***, et al.
   SafeMed-R1: Clinician-Audited Safety and Ethics Alignment for Medical Large Language Models.
   arXiv:2605.28338 arXiv preprint, 2026.
3. **Ding, C.***, Bian, M.*, Yuan, M.*, et al.
   Advancing medical AI through benchmarking and competition for specialty triage. PDF DOI
   npj Digital Medicine, IF: 15.1, 2026.
4. Ding, J.*, **Ding, C.***, Jiang, Y.*, et al.
   Beyond Knowledge to Agency: Evaluating Expertise, Autonomy, and Integrity in Finance with CNFinBench. Project Code
   KDD 2026, accepted. ACM SIGKDD / CCF-A
5. Jiang, Y.*, Li, Q.*, Xu, B.*, Sun, H., **Ding, C**., et al.
   IBISAgent: Reinforcing Pixel-Level Visual Reasoning in MLLMs for Universal Biomedical Object Referring and Segmentation.
   arXiv:2601.03054 CVPR 2026, accepted. IEEE/CVF CVPR / CCF-A
6. **Ding, C.*,** Lu, R.*, Kong, Z., Huang, R.
   TyG index, depression, and cognitive dysfunction: NHANES with machine learning support. PDF DOI
   Journal of Affective Disorders, IF: 4.9, 2025.
7. **Ding, C.***, Yuan, M.*, Cheng, J., Wen, J.
   Smoking types and stroke risk: development of a predictive model for identifying stroke risk. PDF DOI
   Frontiers in Physiology, IF: 3.4, 2025.
8. **Ding, C.,** Kong, Z., Cheng, J., Huang, R.
   U-shaped relationship between TyG index and depression using machine learning. PDF DOI
   Heliyon, IF: 3.6, 2024.

RESEARCH EXPERIENCE

Shanghai Artificial Intelligence Laboratory | Research Assistant *Dec. 2024 – Present*

Project 1: Patient-Facing Multi-agent | *arXiv / Manuscript under review at NEJM AI* | *Dec. 2025– May. 2026*

• Built a simulated patient–doctor evaluation environment integrating patient simulators, frontier LLM doctor backbones, an automated clinical judge, and a SafeMed-R1 in-loop safety controller.
• Formalized delayed escalation as a temporal safety failure and evaluated 5 LLMs across 150 multi-turn consultations, using first-escalation turn and missed-escalation rate against a 40-case clinician reference benchmark.
• Implemented a five-action controller—PASS, REWRITE, ASK-MORE, ESCALATE, REFUSE—reducing mean escalation turn from 6.28 to 3.78 rounds and missed escalation from 62.5% to 12.5%, while revealing GPT-5.1 over-intervention behavior.

Project 2: SafeMed-R1: Medical LLMs | arXiv:2605.28338 *Dec. 2024 – Dec. 2025*

• Developed SafeMed-R1, a safety- and ethics-aligned medical LLM based on Qwen3-32B, using SFT and RL/GRPO with clinician-audited reasoning traces, safety/ethics supervision, and red-team jailbreak stress testing.
• Built the Clinical Trust Signals pipeline for model supervision, including expert-reviewed QA-CoT construction, rubric scoring, adversarial re-answering checks, and benchmark evaluation; achieved 79.6% macro-averaged clinical accuracy and reduced unsafe outputs under adversarial testing.
• Deployed SafeMed-R1 as both a standalone aligned medical model and a modular turn-level governance model for patient-facing dialogue, connecting training-time safety alignment with real-time escalation control. Skills: SFT, RL/GRPO, Qwen-series LLMs, vLLM, red-team evaluation, medical ethics alignment.

Project 3: MedBench / MedTriage: Benchmarking and Specialty Triage | PDF DOI *npj Digital Medicine* / *Dec. 2024 – April.2025*

- Built MedTriage from hospital intake records, online guidance dialogues, and outpatient clinical notes; designed leakage-prevention cleaning and strict multi-label exact-match department recommendation.
- Developed MedGPT-Guide, a retrieval-augmented triage model using BGE-m3 retrieval, 10 relevant + 10 random demonstrations, CoT prompting, candidate-order perturbation, self-consistency voting, and ensemble aggregation; improved accuracy from 0.5319 to 0.7830.
- Contributed to MedBench v4 infrastructure for LLMs, multimodal models, and clinical agents, including rotating test pools, 14 agent-based tasks, task decomposition, tool use, multi-turn planning, safety-critical reasoning, and LLM-as-a-judge calibration.
- Skills: benchmark construction, RAG prompting, clinical data cleaning, multi-label evaluation, clinical agent evaluation, tool-use / workflow assessment, LLM-as-a-judge, medical safety scoring.

Project 4: multi-agent / agentic Benchmark *Dec. 2025 – April.2025*

Expertise, Autonomy, and Integrity Evaluation for High-Stakes Financial LLM Agents | Project Code *KDD 2026, accepted*

- Co-developed CNFinBench, evaluating high-stakes financial agents across expertise, autonomy, and integrity with **29 subtasks**, **11,947** single-turn QA instances, and **321** multi-turn adversarial dialogues.
- Designed agentic workflow evaluation covering requirement parsing, path planning, API / database operations, tool invocation, multi-agent collaboration, and result verification; contributed to Chain-of-Attack evaluation with 17 attacker personas and 7 attack strategies.
- Helped develop HICS to quantify behavioral compliance drift across 22 open- and closed-source models, revealing execution-chain degradation and rapid multi-turn attack collapse.

CLINICAL AND RESEARCH TRAINING

Putuo District Central Hospital, Shanghai University of Traditional Chinese Medicine | Resident Physician *Jul. 2022 – Jul. 2025*

- Completed standardized residency training across emergency medicine, cardiology, respiratory medicine, pediatrics, obstetrics, neurology, and related departments.
- Gained first-hand exposure to red-flag presentations including chest pain, acute dyspnea, obstetric bleeding, pediatric fever, trauma, infection, and neurologic emergencies, which later informed research on AI triage and escalation safety.
- Developed clinical understanding of patient-facing risk communication, urgent referral thresholds, and real-world constraints in early-stage triage.

EDUCATION

Shanghai University of Traditional Chinese Medicine | Master of Medicine, expected Dec 2026

Relevant coursework: Medical Statistics; Research Methodology and Scientific Writing; Immunological Techniques; Medical English; Clinical Practice

Jiangxi University of Traditional Chinese Medicine | Bachelor of Medicine, 2015-2020

Training in anatomy, physiology, pathology, pharmacology, medical imaging, immunology, genetics, biochemistry, microbiology, and cell biology.

HONORS & AWARDS

Provincial	• *3rd Prize*, 18th Challenge Cup Shanghai Collegiate Extracurricular Academic Science & Technology Works Competition	2023/09
University-Wide	• *Yifang Innovation Award, 3rd Prize*	2026/05
	• *Academic Scholarship, Shanghai University of Traditional Chinese Medicine*	2023/09